

Designing Caterpillars for Graphs: Approximation and Hardness^{*}

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Abstract. The classical MINIMUM LINEAR ARRANGEMENT (MLA) problem has been studied extensively. It is known to be NP-hard and it admits an $O(\sqrt{\log n \log \log n})$ -approximation [Feige and Lee, IPL, 2007]. MLA can be defined as follows as design problem: Given a graph G with vertex set $V(G)$, design a path H on the same vertex set that minimizes the linear arrangement cost $\sum_{uv \in E(G)} \text{dist}_H(u, v)$, where $\text{dist}_H(u, v)$ indicates the distance of u and v in H . We initiate the study of the generalization in which H is allowed to be a caterpillar graph of maximum degree at most Δ . Caterpillars are the simplest generalization of paths, having pathwidth one and interpolating between paths and stars via the degree parameter Δ . We give an algorithm that lifts any α -approximation for MLA to an $(\alpha + 3 - 2/(\Delta - 1))$ -approximation for our problem, thus obtaining an $O(\sqrt{\log n \log \log n})$ -approximation for our more general problem as well. Moreover, we derive a 4-approximation whenever MLA is polynomial-time solvable, in particular, for trees. We complement these results by proving NP-hardness for every constant $\Delta \geq 2$ and (in stark contrast to MLA) NP-hardness on trees when Δ is part of the input.

Keywords: Network Design · Minimum Linear Arrangement · Caterpillars

1 Introduction

A *linear arrangement* (also called *linear layout*) of a graph G on n vertices is a bijection $\pi: V(G) \rightarrow [n]$. Linear arrangement problems have been studied for over half a century, motivated by applications ranging from VLSI circuit design and numerical linear algebra to scheduling, graph drawing, and computational biology [5]. There are various objectives that quantify how well the edges of G are preserved by such an arrangement. Three objectives have received particular attention: the *bandwidth* (or *dilation*), $\max_{uv \in E(G)} |\pi(u) - \pi(v)|$; the *cutwidth* (or *congestion*), the maximum number of edges of G crossing any cut of the line; and the *linear arrangement cost*, $\sum_{uv \in E(G)} |\pi(u) - \pi(v)|$. The computational problems

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minimizing the corresponding objective are called BANDWIDTH, CUTWIDTH, and MINIMUM LINEAR ARRANGEMENT (MLA).

All three problems are NP-hard on arbitrary input graphs [9,10,18], and a substantial body of work has mapped out their complexity as one varies the input graph G . When G is a tree, for instance, the picture splits: BANDWIDTH remains NP-hard [8], whereas both CUTWIDTH [24] and MLA [4] admit polynomial-time algorithms. Tractable or well-approximable cases of MLA are known for hypercubes, rectangular grids, dense graphs, and planar graphs, and a long line of work on general graphs has driven the best approximation ratio down to $O(\sqrt{\log n} \log \log n)$ due to Feige and Lee [7].

An equivalent definition can be given from the viewpoint of network design problems: Find a path H with the same vertex set as G that minimizes the respective objective. The three objectives can then be recast as $\max_{uv \in E(G)} \text{dist}_H(u, v)$, the maximum edge-cut over cuts of H , and $\sum_{uv \in E(G)} \text{dist}_H(u, v)$, respectively. From this vantage point it is natural to ask what happens if H can be chosen from a richer class of host graphs? This direction has recently gained traction (see related work). In this paper, we focus on the linear arrangement objective and allow H to be a *caterpillar*, i.e., a tree whose non-leaf vertices induce a path, called the *spine* of the caterpillar, with the remaining leaves (the *legs*) attached to spine vertices. Caterpillars are a natural starting point for at least two reasons. Structurally, they are precisely the connected graphs of pathwidth one [19]. In this sense they form the “closest” generalization of paths. Algorithmically, they give any network design procedure an additional degree of freedom over paths as for each vertex one can decide whether it becomes a spine or a leg.

Note that irrespective of H the cost $\sum_{uv \in E(G)} \text{dist}_H(u, v)$ is always at least $|E(G)|$. Since a star (a caterpillar with one spine vertex) yields a cost of at most $2|E(G)|$ for any G , our problem is trivially 2-approximable. Thus, having great flexibility in the choice of H can lead to better approximation guarantee; this contrasts the non-trivial approximation algorithms for MLA with much higher approximation factor [7]. For a better control of the caterpillar, we extend the input by a bound $\Delta > 1$ on the maximum degree of the desired caterpillar (i.e., we look for a degree- Δ caterpillar). The parameter Δ allows for a fine control on how “MLA-like” H has to be. Formally, we consider the following problem:

GRAPH IN CATERPILLAR (GIC)

Input: An undirected graph G and an integer $\Delta > 1$.

Task: Find a degree- Δ caterpillar H on vertex set $V(G)$ minimizing $\sum_{uv \in E(G)} \text{dist}_H(u, v)$.

Related Work. Hardness of MLA on general graphs was established by Garey, Johnson, and Stockmeyer [9] via a reduction from MAX CUT. For MLA on trees, Goldberg and Klipker [11] gave the first polynomial-time algorithm, with subsequent improvements by Shiloach [23] and Chung [4]. Currently, MLA on an n -vertex tree can be solved in $O(n^{\log 3 / \log 2})$ time. On the approximation side, a line of work [15,6,20] drove the ratio for general graphs from $O(\log^2 n)$ down to $O(\log n)$, and finally to $O(\sqrt{\log n} \log \log n)$ by Feige and Lee [7]. Planar graphs

admit an $O(\log \log n)$ -approximation [20], and dense graphs admit a PTAS [1]. Exact optimal solutions are also known for several structured graph classes, including hypercubes [12] and square grids [17]. See Díaz, Petit, and Serna [5] for further pointers. All of these results restrict H to be a path.

The only prior work we are aware of on varying H falls into two strands. A first line of work considers designing cycles [21] or grids [3] under dilation/congestion measures. Closer to our setting, a second line studies the linear arrangement cost for other host classes. Avin, Mondal, and Schmid [2] gave an $O(1)$ -approximation when H is required to be a degree-4 graph and G is a tree; Kellerhals et al. [13] gave a 4-approximation for the case where H is a binary tree and G is a tree; Schmid et al. [22] gave a polynomial-time algorithm for the case where H is a binary search tree and G is arbitrary; and Liebchen and Wünsch [16] gave an overview on different problem variants where the task is to find a spanning tree H of G under various objectives, including linear arrangement cost.

Our Contribution. We introduce and study G1C as a generalization of MLA (which is G1C with $\Delta = 2$). G1C inherits its NP-hardness from MLA. We extend this NP-hardness to all values $\Delta \geq 2$ (Section 2.2). Somewhat surprisingly, we also extend any approximation result of MLA to G1C (Section 2.1): Our algorithm takes any α -approximate linear arrangement and rounds it into a degree- Δ caterpillar whose cost is within an additive $O(|E(G)|)$ term of α times the optimum. This implies that for any graph class for which MLA is polynomial-time solvable (e.g., trees), G1C admits a polynomial-time 4-approximation algorithm.

We also discover differences in the complexity of MLA and G1C: We show that G1C remains NP-hard if G is a tree (Section 3), whereas MLA is polynomial-time solvable in this case. This hints that trying to transfer polynomial-time (exact) algorithms from MLA to G1C might be a pointless endeavor.

Preliminaries. All graphs are assumed to be undirected. For a given graph G , we denote its vertices by $V(G)$ and its edges by $E(G)$. For $u, v \in V(G)$, we denote by $\text{dist}_G(u, v)$ the length of a shortest uv -path if one exists and infinity otherwise. The *cost* of a graph H with $V(H) = V(G)$ is

$$\text{cost}(G, H) := \sum_{uv \in E(G)} \text{dist}_H(u, v).$$

We say an edge $uv \in E(G)$ is *stretched* by H if $\text{dist}_H(u, v) - 1 > 0$. For a graph class $\mathcal{H}$, we define $\text{opt}_{\mathcal{H}}(G) := \min\{\text{cost}(G, H) \mid H \in \mathcal{H}, V(G) = V(H)\}$, and we say that $H \in \mathcal{H}$ is *optimal* if $V(G) = V(H)$ and the cost of H is $\text{opt}_{\mathcal{H}}(G)$.

A *caterpillar* is a tree C whose non-leaf vertices induce a path, called the *spine*. A leaf u in C is called *leg* of spine vertex v , where v is the unique neighbor of u . For a spine vertex v , the *segment* $\text{seg}_C(v)$ of v is the subgraph of C induced by v and all legs of v . We write $\text{seg}(v)$ if C is clear from the context. A caterpillar is a *degree- Δ caterpillar* if it has maximum degree at most Δ . A caterpillar is *Δ -legged* if each of its segments has at most Δ legs. Note that the only difference between restricting the degree of a caterpillar and restricting the number of legs

a segment can have is whether the outermost segments are allowed to have one more leg than the inner segments or not.

2 Designing Caterpillars for Graphs

In this section, we first show our main result: an $O(\sqrt{\log n} \log \log n)$ -approximation for GIC which is based on the known approximation for MLA [7]. Our approximation works for any given degree bound Δ for the caterpillar to be found. We subsequently show NP-hardness of GIC for any given constant Δ .

2.1 Approximation Algorithm

We next show that any α -approximation for the MLA problem can be lifted to an $(\alpha + 3 - 2/(\Delta - 1))$ -approximation for GIC. Combined with the $O(\sqrt{\log n} \log \log n)$ -approximation for MLA due to Feige and Lee [7], this yields an approximation of the same order for our problem.

For $\Delta = n - 1$, a star already serves as a trivial 2-approximation, since every edge of G is stretched to a path of length at most 2 and the optimum is at least $|E(G)|$. Our algorithm extends this idea to arbitrary $\Delta \geq 2$: it builds stars of size $\Delta - 1$ and concatenates them along a spine. The stars are dictated by a linear arrangement π of G : the i -th star contains those vertices placed by π into positions $(i - 1)(\Delta - 1) + 1, \dots, i(\Delta - 1)$. For each star, its center is chosen to be a vertex of maximum degree in G .

We make this formal as follows. A linear arrangement of G is a bijection $\pi: V(G) \rightarrow [n]$. For a linear arrangement π and a vertex v , let

$$\beta_\pi(v) := \left\lceil \frac{\pi(v)}{\Delta - 1} \right\rceil$$

be the index of the star that v belongs to.

STARFY algorithm. Given a linear arrangement π , proceed as follows.

1. Partition $V(G)$ into $\lceil n/(\Delta - 1) \rceil$ consecutive segments $S_i := \{v \in V(G) \mid \beta_\pi(v) = i\}$.
2. For each segment S_i , choose a vertex $s_i \in S_i$ of maximum degree in G and turn S_i into a star centered at s_i .
3. Connect the centers along a spine by adding the edges $s_{i-1}s_i$ for all $i > 1$.

We analyze STARFY in two steps. First, we bound the cost of the caterpillar H produced by STARFY in terms of the cost of the input arrangement π (Lemma 1). Second, we bound the cost of an MLA from above by the cost of a path obtained by *flattening* an optimal caterpillar (Lemma 2). Combining the two bounds yields the main result (Theorem 1).

Lemma 1. *Let π be a linear arrangement of G , and let H be the caterpillar produced by STARFY on input π . Then*

$$\sum_{uv \in E(G)} \text{dist}_H(u, v) \leq \sum_{uv \in E(G)} |\beta_\pi(u) - \beta_\pi(v)| + \left(2 - \frac{2}{\Delta - 1}\right) |E(G)|.$$

Proof. Let $L := V(G) \setminus \{s_1, s_2, \dots\}$ be the set of legs of H , that is, the vertices that are not the center of their segment. Fix an edge $uv \in E(G)$. In H , the unique uv -path decomposes into three pieces: from u to the center $s_{\beta_\pi(u)}$ of its segment, along the spine from $s_{\beta_\pi(u)}$ to $s_{\beta_\pi(v)}$, and from $s_{\beta_\pi(v)}$ to v . The first and third pieces have length at most 1, and length 0 precisely when u (respectively v) is the center of its segment. The middle piece has length $|\beta_\pi(u) - \beta_\pi(v)|$. Hence

$$\text{dist}_H(u, v) \leq |\beta_\pi(u) - \beta_\pi(v)| + |\{u, v\} \cap L|.$$

Summing over all edges $uv \in E(G)$,

$$\sum_{uv \in E(G)} |\{u, v\} \cap L| = \sum_{v \in L} \deg_G(v) = 2|E(G)| - \sum_i \deg_G(s_i).$$

Since each center s_i is chosen to have maximum degree in its segment S_i ,

$$\sum_i \deg_G(s_i) \geq \sum_i \frac{1}{|S_i|} \sum_{v \in S_i} \deg_G(v) \geq \frac{1}{\Delta - 1} \sum_{v \in V(G)} \deg_G(v) = \frac{2|E(G)|}{\Delta - 1}.$$

Combining these two displays yields

$$\sum_{uv \in E(G)} \text{dist}_H(u, v) \leq \sum_{uv \in E(G)} |\beta_\pi(u) - \beta_\pi(v)| + 2|E(G)| - \frac{2|E(G)|}{\Delta - 1}. \quad \square$$

To bound the cost of an MLA, we bound the cost of a specific linear arrangement obtained by linearizing an optimal caterpillar.

FLATTEN algorithm. Given a caterpillar H with spine $s_1 \dots s_k$ (in some fixed left-to-right order), construct a path H' on $V(H)$ as follows: for each spine vertex s_i with $i < k$, insert the legs of s_i immediately to the left of s_i (in arbitrary order); insert the legs of s_k immediately to the right of s_k . The result is a path in which the spine vertices appear in their original left-to-right order, each preceded by its legs (except for s_k , which is followed by its legs).

Lemma 2. *Let H be a degree- Δ caterpillar, and let H' be the path obtained from H via FLATTEN. Then for every pair of vertices $u, v \in V(H)$,*

$$\text{dist}_{H'}(u, v) \leq (\Delta - 1) \cdot \text{dist}_H(u, v).$$

Proof. Let $P = v_0 v_1 \dots v_\ell$ be the unique uv -path in H , with $v_0 = u$ and $v_\ell = v$, oriented from left to right with respect to the orientation fixed by FLATTEN. (If

u and v land in the same spine segment, orient arbitrarily.) After FLATTEN, the vertices $v_0, v_1, \dots, v_\ell$ appear in this same left-to-right order along H' , possibly with additional legs of $v_1, \dots, v_\ell$ interspersed between consecutive vertices of P . For each $i \in [\ell]$, if v_i is an interior spine vertex, it has at most $\Delta - 2$ legs (since it has two spine neighbors); if v_i is the rightmost spine vertex, FLATTEN places its legs to its right, so none are inserted before v_i ; and if v_i is a leg of H , it has no legs of its own. In every case, at most $\Delta - 2$ vertices are inserted between v_{i-1} and v_i . (The legs of $v_0 = u$ are placed to the left of v_0 by FLATTEN and therefore do not appear between v_0 and v_ℓ .) Summing over all ℓ steps, we have

$$\text{dist}_{H'}(u, v) \leq \ell + (\Delta - 2)\ell = (\Delta - 1) \cdot \text{dist}_H(u, v). \quad \square$$

We now combine the two lemmas.

Theorem 1. *Let H^* be an optimal degree- Δ caterpillar for G , let π be an α -approximation for MLA on G , and let H be the caterpillar produced by STARFY on input π . Then*

$$\sum_{uv \in E(G)} \text{dist}_H(u, v) \leq \alpha \sum_{uv \in E(G)} \text{dist}_{H^*}(u, v) + \left(3 - \frac{2}{\Delta - 1}\right) |E(G)|.$$

Proof. By Lemma 1 and the bound $|\beta_\pi(u) - \beta_\pi(v)| \leq |\pi(u) - \pi(v)|/(\Delta - 1) + 1$,

$$\begin{aligned} \sum_{uv \in E(G)} \text{dist}_H(u, v) &\leq \sum_{uv \in E(G)} \left(\frac{|\pi(u) - \pi(v)|}{\Delta - 1} + 1 \right) + \left(2 - \frac{2}{\Delta - 1}\right) |E(G)| \\ &= \frac{1}{\Delta - 1} \sum_{uv \in E(G)} |\pi(u) - \pi(v)| + \left(3 - \frac{2}{\Delta - 1}\right) |E(G)|. \end{aligned}$$

Let π^* be an MLA of G . Since π is an α -approximation,

$$\sum_{uv \in E(G)} \text{dist}_H(u, v) \leq \frac{\alpha}{\Delta - 1} \sum_{uv \in E(G)} |\pi^*(u) - \pi^*(v)| + \left(3 - \frac{2}{\Delta - 1}\right) |E(G)|.$$

Let $(H^*)'$ be the path obtained from H^* via FLATTEN, and let π' be its induced linear arrangement. Since π^* is optimal,

$$\sum_{uv \in E(G)} |\pi^*(u) - \pi^*(v)| \leq \sum_{uv \in E(G)} |\pi'(u) - \pi'(v)| = \sum_{uv \in E(G)} \text{dist}_{(H^*)'}(u, v).$$

By Lemma 2, $\text{dist}_{(H^*)'}(u, v) \leq (\Delta - 1) \text{dist}_{H^*}(u, v)$ for every $uv \in E(G)$, hence

$$\begin{aligned} \sum_{uv \in E(G)} \text{dist}_H(u, v) &\leq \frac{\alpha}{\Delta - 1} \sum_{uv \in E(G)} (\Delta - 1) \text{dist}_{H^*}(u, v) + \left(3 - \frac{2}{\Delta - 1}\right) |E(G)| \\ &= \alpha \sum_{uv \in E(G)} \text{dist}_{H^*}(u, v) + \left(3 - \frac{2}{\Delta - 1}\right) |E(G)|. \quad \square \end{aligned}$$

The trivial lower bound $\sum_{uv \in E(G)} \text{dist}_{H^*}(u, v) \geq |E(G)|$ in Theorem 1 yields

$$\sum_{uv \in E(G)} \text{dist}_H(u, v) \leq \left(\alpha + 3 - \frac{2}{\Delta - 1} \right) \sum_{uv \in E(G)} \text{dist}_{H^*}(u, v),$$

which is what we invoke in Corollaries 1 and 2. The bound of Theorem 1 however degrades gracefully with the optimum: when $\sum_{uv \in E(G)} \text{dist}_{H^*}(u, v) = \omega(|E(G)|)$, the additive term is $o(1)$, and STARFY is an $(\alpha + o(1))$ -approximation.

The additive slack also vanishes at the extreme value $\Delta = 2$: here STARFY returns the input linear arrangement unchanged. Since $\Delta - 1 = 1$ divides n , the ceiling terms in the proof of Lemma 1 contribute no rounding, so the approximation factor reduces to α .

The best known approximation for MLA [7] implies the following.

Corollary 1. *For every $\Delta \geq 2$, there is an $O(\sqrt{\log n} \log \log n)$ -approximation algorithm for GRAPH IN CATERPILLAR.*

Proof. Run the $O(\sqrt{\log n} \log \log n)$ -approximation algorithm for MLA of Feige and Lee [7] to obtain a linear arrangement π , and apply STARFY. Theorem 1 and the lower bound $\sum_{uv \in E(G)} \text{dist}_{H^*}(u, v) \geq |E(G)|$ imply the result. $\square$

Plugging in known (approximation) algorithms for MLA on restricted graph classes, we obtain the following specific ratios.

Corollary 2. *STARFY yields a polynomial-time α -approximation algorithm for GIC with*

- $\alpha = 4 - 2/(\Delta - 1)$, when G is a tree, a hypercube, or a square grid; and
- $\alpha = O(\log \log n)$, when G is planar.

Proof. MLA is polynomial-time solvable for trees [4], hypercubes [12], and square grids [17] and admits a polynomial-time $O(\log \log n)$ -approximation for planar graphs [20]. Invoking Theorem 1 with the lower bound $\sum_{uv \in E(G)} \text{dist}_{H^*}(u, v) \geq |E(G)|$ yields the claimed approximation algorithm for GIC. $\square$

2.2 NP-hardness

In this section, we show that the problem of designing optimally degree- Δ caterpillars for arbitrary graphs is NP-hard for every constant $\Delta \geq 2$. For $\Delta < 2$, this problem becomes trivial. We generalize the NP-hardness proof of Garey et al. [9] for MINIMUM LINEAR ARRANGEMENT via a reduction from MAX CUT, where, given a graph $G = (V, E)$ and $k \in \mathbb{N}$, the task is to find a set $S_1 \subseteq V$ such that there are at least k edges between S_1 and $S_2 := V \setminus S_1$.

We first sketch the NP-hardness proof of MLA by Garey et al. [9]. Given a MAX CUT instance $(G = (V, E), k)$, let G' be the complement of G augmented by a set U of $r = n^4$ vertices adjacent to all other vertices, and set $W = \text{opt}(K_{r+n}) - kr$, where $\text{opt}(K_{r+n})$ is the cost of an optimal embedding of K_{r+n} into a path.

Minimizing the cost then amounts to placing the endpoints of the non-edges of G' —that is, the edges of G —as far apart as possible on the path, while the n^4 vertices of U outweigh any cost contribution of edges within one part. Consequently, an optimal arrangement can be assumed to place all of U consecutively, with no $v \in V$ having vertices of U both to its left and to its right. The two sides yield a partition $S_1 \cup S_2 = V$ whose cut has size at least k . Our reduction follows the same scheme with two changes: we swap whole segments instead of single vertices, and we additionally bound the maximum distance between two vertices of the caterpillar—without such a bound, the resulting lower bound on the cut size is too weak.

Theorem 2 ($\star^3$). *GRAPH IN CATERPILLAR is NP-hard for each fixed $\Delta \geq 2$.*

3 Designing Caterpillars for Trees

Recall that our general approximation approach provides a 4-approximation for GIC if the input graph is a tree (see Corollary 2); we will subsequently refer to this special case of GRAPH IN CATERPILLAR as TREE IN CATERPILLAR (TIC). An immediate question is, whether turning to approximation to solve TIC is necessary? While the polynomial-time algorithms for MLA for trees are non-trivial, the structure of minimum linear arrangements of trees is by now well-understood [4,11,23]. In preliminary considerations, we were able to transfer a few of these structural observations, but not all. Somewhat surprisingly, we cannot transfer all observations as we prove TIC to be NP-hard.

A natural question is why the techniques underlying the tractability of MLA on trees do not carry over to caterpillar hosts. All three mentioned algorithms for MLA on trees follow a divide-and-conquer approach: they decompose the input tree into subtrees, recursively compute MLAs for the subtrees, and concatenate the resulting arrangements into an MLA for the whole tree. Crucially, this strategy relies on the property that an MLA for the whole tree, restricted to any subtree in the decomposition, is again an MLA for that subtree. This property fails for caterpillar hosts. While an optimum caterpillar for a path-like subtree may again be path-like in isolation, in the context of the whole tree it can be beneficial to *compress* such a caterpillar—that is, to move vertices off the spine and attach them as legs—to shorten edges with one endpoint in the subtree and one endpoint outside. Consequently, an optimum caterpillar for the whole tree need not restrict to an optimum caterpillar for a subtree, which obstructs a direct divide-and-conquer approach.

We reduce from the strongly NP-hard problem 3-PARTITION (see §4.2.2, Theorem 4.4 in the book by Garey and Johnson [8]).

3-PARTITION

Input: Positive integers $a_1, \dots, a_{3n}$ and $B \in \mathbb{N}$ satisfying $B/4 < a_i < B/2$.

Task: Is there a partition of the index set $\{1, \dots, 3n\}$ into parts $P_1, \dots, P_n$ such that $|P_j| = 3$ and $\sum_{i \in P_j} a_i = B$ for all $1 \leq j \leq n$?

³ Proofs of statements marked with $\star$ are deferred to the full version [14].

The general idea of our reduction is straightforward. We construct a tree T that contains $3n$ disjoint subtrees $A_1, \dots, A_{3n}$ of sizes $a_1, \dots, a_{3n}$, respectively, and n subtrees $B_1, \dots, B_n$, each of size $\kappa_2 + 1$ (here κ_2 is a number much larger than B). We choose the degree bound to be $\Delta = \kappa_2 + B$. The goal is to adapt T and W so that any caterpillar C with $\text{cost}(T, C) \leq W$ will have n segments containing all vertices from one subtree B_j and three subtrees $A_{j_1}, A_{j_2}, A_{j_3}$. Thus, any partition of the index set satisfying the 3-PARTITION conditions will correspond to the segments of a caterpillar C with $\text{cost}(T, C) = W$ and vice versa.

Most of the proof deals with the construction of T . In order to force the roots (called $\beta_1, \dots, \beta_n$ in the proofs) of $B_1, \dots, B_n$ to all be spine vertices (and thus not appear in the same segment), we give them so many legs that if we would not choose them as spine vertices, the cost would exceed W . A simple swapping argument then gives that their legs must also be in the same segment, that is, each B_j is a subtree of a segment. As we want the vertices of $A_1, \dots, A_{3n}$ to be legs of these segments as well, we need to exclude the possibility of them forming segments themselves. This is achieved by introducing high-degree subtrees (called $B_1^*, \dots, B_{\kappa_1}^*$ in the proofs) to T which each induces a segment of C for the same reasons $B_1, \dots, B_n$ do. Then, forming a segment solely for some A_i , would make the spine too long and again the cost would exceed W . Hence, C can be assumed to have the intended structure mentioned above together with additional segments for $B_1^*, \dots, B_{\kappa_1}^*$. Consult Figure 1 and Figure 2 for visualizations of the tree T and the (intended) caterpillar C , respectively. The bound W is chosen as the cost of the intended caterpillar for T .

Theorem 3. *TREE IN CATERPILLAR is NP-hard.*

Instead of proving this theorem directly, we opt for the more convenient proof where Δ is an upper bound on the legs of the caterpillar instead of an upper bound on its degree, that is, we show NP-hardness for the problem of—given a tree T and $\Delta, W \in \mathbb{N}$ —finding a Δ -legged caterpillar C for T with cost at most W . This avoids discussing the edge cases for the outermost segments which can have one more leg than the inner segments if the degree is bounded. Later, we remark how this proof implies NP-hardness for TiC.

To prove this, let us begin by constructing T and choosing $\Delta, W \in \mathbb{N}$. Suppose integers $a_1, \dots, a_{3n}$ and B for some $n \in \mathbb{N}$ such that $B/4 < a_i < B/2$ are given. We assume without loss of generality that $a_1 + \dots + a_{3n} = nB$; otherwise, a desired partition of the index set can not exist. Define

$$\sigma(n) := \begin{cases} \frac{(n-1)^2}{4} - \frac{n-1}{2} & \text{if } n \text{ is odd} \\ \frac{(n-1)^2}{4} & \text{if } n \text{ is even} \end{cases} \quad \text{and} \quad \lambda(n) := 3(\sigma(n) + n - 1)$$

and define the following numbers (whose purpose will be explained later):

$$\begin{aligned} \kappa_0 &:= nB, & \kappa_1 &:= 2((\kappa_0 - 3n) + \lambda(n) + 1) + 1, \\ \kappa_2 &:= (\kappa_0 - 3n) + \sigma(n + \kappa_1) + \lambda(n) + B + 1. \end{aligned}$$

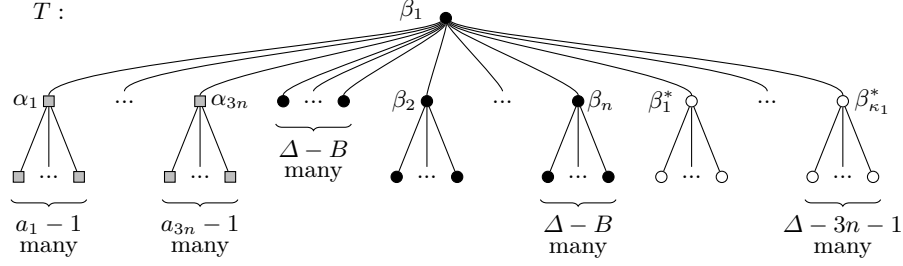


Fig. 1. The tree T constructed in our reduction.

We will first construct the tree T . For all $1 \leq i \leq 3n$, let A_i be a new tree of depth 1 with root α_i and $a_i - 1$ leaves. For all $1 \leq j \leq n$, let B_j be a new tree of depth 1 with root β_j and κ_2 leaves. For all $1 \leq k \leq \kappa_1$, let B_k^* be a new tree of depth 1 with root β_k^* and $\kappa_2 + B - 3n - 1$ leaves. We will call the leaves of A_i , B_j and B_k^* α_i -leaves, β_j -leaves and β_k^* -leaves, respectively. We now obtain T by connecting β_1 with $\alpha_1, \dots, \alpha_{3n}$, $\beta_2, \dots, \beta_n$ and $\beta_1^*, \dots, \beta_{\kappa_1}^*$, that is,

$$\begin{aligned} V(T) &:= \bigcup_{i=1}^{3n} V(A_i) \cup \bigcup_{j=1}^n V(B_j) \cup \bigcup_{k=1}^{\kappa_1} V(B_k^*), \\ E(T) &:= \bigcup_{i=1}^{3n} E(A_i) \cup \bigcup_{j=1}^n E(B_j) \cup \bigcup_{k=1}^{\kappa_1} E(B_k^*) \cup \{\beta_1 \alpha_i \mid 1 \leq i \leq 3n\} \\ &\quad \cup \{\beta_1 \beta_j \mid 2 \leq j \leq n\} \cup \{\beta_1 \beta_k^* \mid 1 \leq k \leq \kappa_1\}. \end{aligned}$$

Consult Figure 1 for a visualization of the tree T . We define

$$W := |E(T)| + \sigma(n + \kappa_1) + \lambda(n) + (\kappa_0 - 3n) \quad \text{and} \quad \Delta := \kappa_2 + B.$$

The bound W will turn out to be exactly the cost of the intended caterpillar C for T constructed in the proof of Lemma 3 which is visualized in Figure 2.

We now prove the forward direction of the reduction, that is, for a given 3-PARTITION YES-instance, we explicitly give a Δ -legged caterpillar C such that $\text{cost}(T, C) = W$.

Lemma 3. *If there is a partition of $\{1, \dots, 3n\}$ into parts $P_1, \dots, P_n$ such that $P_j = \{j_1, j_2, j_3\}$ and $a_{j_1} + a_{j_2} + a_{j_3} = B$ for all $1 \leq j \leq n$, then there is a Δ -legged caterpillar C with $\text{cost}(T, C) = W$.*

Proof. We construct a caterpillar C with $V(C) = V(T)$ such that $\text{cost}(T, C) = W$. Its spine $s_1 \dots s_\ell$ will consist of the β_j and β_k^* . Thus, the spine will be of length $\ell = n + \kappa_1$. On the spine, β_1 is placed in the “middle”, that is, $\beta_1 = s_{\lceil \ell/2 \rceil}$. Then, the β_j are arranged as close as possible to β_1 , that is,

$$\{\beta_2, \dots, \beta_n\} = \{s_k \mid \lceil \ell/2 \rceil - \lfloor (n-1)/2 \rfloor \leq k \leq \lceil \ell/2 \rceil + \lceil (n-1)/2 \rceil\} \setminus \{s_{\lceil \ell/2 \rceil}\}.$$

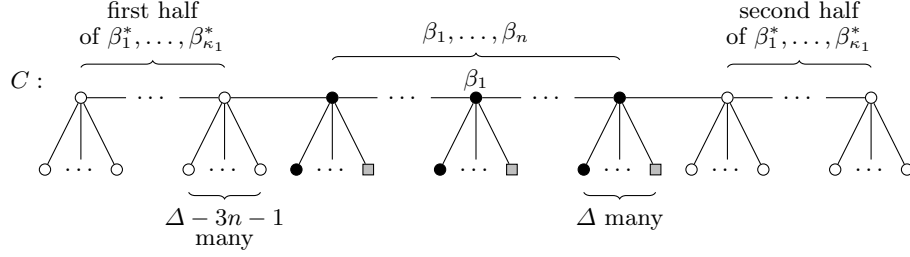


Fig. 2. The intended caterpillar C for T . Each segment with spine vertex β_j consists of the subtree B_j together with three subtrees $A_{j_1}, A_{j_2}, A_{j_3}$ (square vertices) and their leaves; each segment with spine vertex β_k^* consists of B_k^* .

The β_k^* can be placed arbitrarily as the rest of the spine vertices. For each $1 \leq k \leq \kappa_1$, the β_k^* -leaves are the legs of β_k^* . For $1 \leq j \leq n$, the β_j -leaves are legs of β_j . Lastly, for every part $P_j = \{j_1, j_2, j_3\}$, $1 \leq j \leq n$, of the partition of indices of the 3-PARTITION YES-instance, place $\alpha_{j_1}, \alpha_{j_2}, \alpha_{j_3}$ together with their respective leaves as legs of some $\beta_{j'}$ (whose segment does not contain any α_i yet). Now, each spine vertex has at most Δ many legs. Consult Figure 2 for a visualization of C . It remains to show that $\text{cost}(T, C) = W$. To see this, first observe that the cost $|E(T)|$ is inevitable for any caterpillar C . As the edges between some α_i to any of its leaves get stretched to paths of length 2, they induce additional cost 1. There are $\sum_{i=1}^{3n} (a_i - 1) = \kappa_0 - 3n$ such edges. The cost $\sigma(n + \kappa_1)$ arise from the stretching of edges of the form $\beta_1\beta_j$ and $\beta_1\beta_k^*$; they are Gauss sums over the distances from β_1 to all the β_j, β_k^* on the left and on the right of β_1 , respectively. Note that they do not include the cost already covered by the summand $|E(T)|$ and that they are sensitive to the position of β_1 —and thus also to the parity of $n + \kappa_1$. It is now easy to check that these cost are precisely $\sigma(n + \kappa_1)$. Lastly, consider the cost induced by edges of the form $\beta_1\alpha_i$. As the α_i appear as triples as legs of some β_j and the cost induced by the α_j that are β_1 -legs are covered by $|E(T)|$, we get the additional cost of $\lambda(n)$. The edges between β_j, β_k^* and their leaves do not get stretched. Thus, $\text{cost}(T, C) = W$. $\square$

Before proving the converse direction, we state two lemmas about optimal caterpillars for graphs. The first one shows that the spine vertex of a segment has maximum degree among all the vertices in that segment.

Lemma 4 (Spine-Degree-Lemma; $\star$). *Let G be a graph and let C be an optimal caterpillar for G . Let S be a segment of C with spine node $s \in V(S)$. Then, the following holds:*

- $\deg_G(s) = \max\{\deg_G(v) \mid v \in V(S)\}$, and
- swapping s with $v \in V(S)$ for which $\deg_G(s) = \deg_G(v)$ holds also yields an optimal caterpillar for G .

The second lemma characterizes minimum linear arrangements of star graphs up to isomorphism.

Lemma 5 (Embedding Stars into Paths; $\star$). *Let $S = (V, E)$ be a star, that is, $V = \{c\} \cup \{v_1, \dots, v_{n-1}\}$ and $E = \{cv_i \mid 1 \leq i \leq n-1\}$. Then, the optimal embeddings of S into a path $p_1 \dots p_n$ are the embeddings where*

$$c = \begin{cases} p_{\lceil n/2 \rceil} & \text{if } n \text{ is odd,} \\ p_{n/2} \text{ or } p_{n/2+1} & \text{if } n \text{ is even.} \end{cases}$$

Now, we are ready to prove the converse direction. We show that if there is a Δ -legged caterpillar C with $\text{cost}(T, C) \leq W$, then we can assume C to have the same structure as in the forward direction. This allows us to determine a partition of the index set from legs of specific segments. Moreover, the cost must be tight, that is, $\text{cost}(T, C) = W$.

Lemma 6. *If T is constructed and $\Delta, W \in \mathbb{N}$ are chosen from a given 3-PARTITION instance $\{a_1, \dots, a_{3n}\}$ and B as above and there is a caterpillar C such that $\text{cost}(T, C) \leq W$, then there is a partition of $\{1, \dots, 3n\}$ into parts $P_1, \dots, P_n$ such that $P_j = \{j_1, j_2, j_3\}$ and $a_{j_1} + a_{j_2} + a_{j_3} = B$ for all $1 \leq j \leq n$. Moreover, $\text{cost}(T, C) = W$.*

Proof. Let (T, W, Δ) be as constructed above and let C be a Δ -legged caterpillar with $V(C) = V(T)$ and $\text{cost}(T, C) \leq W$. We sketch why $a_1, \dots, a_{3n}$ and B is then a YES-instance of 3-PARTITION; the details are given in the full version [14]. By scaling $a_1, \dots, a_{3n}$ and B we may assume without loss of generality that (1) $B \geq 3n + 1$, (2) $a_i \geq \lambda(n) + 3n + 4$, and (3) $a_i \geq B/4 + \lambda(n) + 1$, where we get the third assumption by scaling with $4(\lambda(n) + 1)$, as initially $a_i \geq B/4 + 1/4$. Let $\mathcal{B} = \{\beta_1, \dots, \beta_n, \beta_1^*, \dots, \beta_{\kappa_1}^*\}$.

- (a) No two $\beta, \beta' \in \mathcal{B}$ are in the same segment of C : the subtrees of β and β' contain $2\kappa_2 + 2$ vertices, at most $\kappa_2 + B + 1$ of which fit into a single segment, so at least $\kappa_2 - B + 1$ of their leaves are in other segments and each of these incurs additional cost at least one, exceeding W by the choice of κ_2 . Together with the Spine-Degree-Lemma (Lemma 4) we may thus assume that all $\beta \in \mathcal{B}$ are spine vertices of C .
- (b) We may further assume that for each $\beta \in \mathcal{B}$ all β -leaves are in the segment S containing β : a β -leaf outside of S can be swapped with a leg of S that is not a β -leaf without increasing the cost, and by (1) such a leg exists.
- (c) No α_i is a spine vertex: otherwise, by the Spine-Degree-Lemma, the neighbors of β_1 occupy at least $n + \kappa_1 + 1$ segments, and Lemma 5 together with the choice of κ_1 yields $\text{cost}(T, C) > W$.
- (d) No α_i is a leg of some β_k^* : by (b) and the choice of Δ , at most $3n + 1$ vertices of A_i are legs of that segment, so by (2) at least $\lambda(n) + 3$ many α_i -leaves are elsewhere, which again gives $\text{cost}(T, C) > W$.
- (e) At most three α_i are in the same segment: by (3), four subtrees A_i together contain at least $B + 4\lambda(n) + 4$ vertices, of which at most B are legs of that segment, and the leaves left outside push the cost above W . Together with (c), (d), and the fact that there are exactly n segments with a spine vertex β_j , exactly three α_i are in each such segment.

- (f) All α_i -leaves are in the segment containing α_i : the edges from the α_i to their leaves incur additional cost $\kappa_0 - 3n$ only if this is the case, and by (a)–(e) we already have cost at least $|E(T)| + \sigma(n + \kappa_1) + \lambda(n)$.

Hence each segment S with spine vertex β_j contains all κ_2 many β_j -leaves and exactly three subtrees $A_{j_1}, A_{j_2}, A_{j_3}$ in its remaining at most B leg positions. As $a_1 + \dots + a_{3n} = nB$, this forces $a_{j_1} + a_{j_2} + a_{j_3} = B$, and we read off the parts $P_j := \{j_1, j_2, j_3\}$ from the legs of $\text{seg}(\beta_1), \dots, \text{seg}(\beta_n)$. Moreover, this forces β_1 to be in the “middle” of the spine with $\beta_2, \dots, \beta_n$ arranged as close as possible around it, so C has the form of the caterpillar constructed in Lemma 3 and thus $\text{cost}(T, C) = W$. $\square$

To complete the proof of Theorem 3 (in the Δ -legged case), observe that we can construct the instance (T, W, Δ) from the 3-PARTITION instance $a_1, \dots, a_{3n}$ and B in polynomial time as 3-PARTITION is strongly NP-hard.

Remark 1. To see why Theorem 3 also holds if we bound the degree instead of the maximal number of legs, we set the degree bound to $\Delta + 2$. Then, carefully scanning the proof above shows that they almost hold for the degree bounded case as well. The only way the proof breaks is when the 3-PARTITION instance contains $a_{i_1}, a_{i_2}, a_{i_3}$ such that $a_{i_1} + a_{i_2} + a_{i_3} = B + 1$. Then, we could give $\alpha_{i_1}, \alpha_{i_2}, \alpha_{i_3}$ and all but one of their leaves as legs to an outermost spine vertex s . The remaining leaf is then added as a new spine vertex next to s on the spine. This breaks step (f) in the proof above. However, we can easily circumvent this edge case by scaling the 3-PARTITION instance by 2. Then, $a_{i_1} + a_{i_2} + a_{i_3}$ is even and thus $a_{i_1} + a_{i_2} + a_{i_3} \neq B + 1$.

Remark 2. Notably, the above proof also shows NP-hardness for a variant of TREE IN CATERPILLAR where the caterpillar C' is given in the input and the task is to find a bijection $\pi : V(T) \rightarrow V(C')$ that minimizes the cost function $\sum_{uv \in E(T)} \text{dist}_{C'}(\pi(u), \pi(v))$. Now, choosing T as above and C' isomorphic to C yields the desired reduction from 3-PARTITION.

4 Conclusion

We studied GIC, a generalization of minimum linear arrangement (MLA), where the goal is to design a caterpillar H with $V(H) = V(G)$ and maximum degree Δ (instead of a path) that minimizes $\sum_{uv \in E(G)} \text{dist}_H(u, v)$. On the one side, we show that we can lift any α -approximation for MLA to an $(\alpha + 3 - 2/(\Delta - 1))$ -approximation for GIC. On the other side the problem remains NP-hard when $\Delta \geq 2$ is constant, and when G is a tree. The latter result depicts a stark contrast to MLA, which is polynomial-time solvable when G is a tree [4]. We conclude with some future research directions.

First, can the approximation ratios be improved (or complemented by lower bounds)? For general graphs, any approximation better than $O(\sqrt{\log n \log \log n})$ would rely on an entirely new approach. For trees, the situation seems more

hopeful: MLA is polynomial-time solvable here [4], so the constant 4 comes entirely from the additive $\Theta(|E(G)|)$ slack in STARFY, which vanishes on instances where the optimum is $\omega(|E(G)|)$. A tailored algorithm exploiting the tree structure of G directly might yield a smaller constant.

The second research direction revolves around the degree bound Δ . Is GtC on trees fixed-parameter tractable with respect to Δ ? And what happens, if we drop the degree bound? Then we would always be “allowed” to choose a star as our caterpillar; however this need not be optimal. The complexity of this case is left open, even if G is a tree.

Looking at the bigger picture, one could study the problem of designing caterpillars for graphs under other objectives, such as the initially mentioned bandwidth or cutwidth measures. Finally, considering other classes of graphs (as has been done in previous work [21,3,2,22,13]) may be a fruitful research avenue.

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